GG-B-001275(DSA-DM)
17 October 1967
Interim revision of
Fed. Spec. GG-B-821 (in part)
14 May 1948

INTERIM FEDERAL SPECIFICATION

BURS, DENTAL, FINISHING

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-B-821 (see 6.3).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers finishing burs, suitable for use in finishing dental restorations.

1.2 Classification.

1.2.1 Classes. Finishing burs shall be of the following classes (see 6.1):

Class AL. For use with latch type, angle handpiece.
Class S. For use with straight handpiece.

1.2.2 Shapes and size numbers. Finishing burs shall be of the following shapes and size numbers:

Interproximal	No. 6
Round	No. 200
Oval	No. 218
Bud	No. 224
Sugar-loaf	No. 242

FSC 6520

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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SPECIFICATIONS

Military

MIL-C-16173 Corrosion Preventive Compound, Solvent Cutback, Cold-Application.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The burs shall be fabricated from steel having a minimum tungsten content of 3.00 percent, and a minimum carbon content of 1.00 percent. As an alternate, the burs shall be fabricated from steel having a minimum tungsten content of 1.00 percent, a minimum vanadium content of 0.08 percent and a minimum carbon content of 1.00 percent.

3.2 Hardness. The bur heads shall conform to the Scratch test requirements of 3.2.1. If the Scratch test fails, or the results are not conclusive, the Vickers test shall be made, and the requirements of 3.2.2 shall have precedence. The bur necks shall be properly tempered to resist fracture or bending in use.

3.2.1 Scratch test. When tested in accordance with 4.4.2, the bur heads shall scratch the plate glass, and the blades shall show no evidence of fracture or deformation (such as turning back) as a result of the testing.

3.2.2 Vickers test. When tested in accordance with 4.4.3, the hardness of the bur heads shall be no less than number (No.) 750 on the Vickers scale (Diamond pyramid hardness).

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3.3 Style, design, and dimensions. The burs shall be dental finishing burs, suitable for use with straight or angle handpieces, as applicable, in finishing dental restorations. The burs shall consist of a shank, neck and head section.

3.3.1 Shank. The shank dimensions and the overall length of the bur shall be as specified in table I for the respective classes.

Table I. Shank dimensions and bur length

Class	Diameter of shank, inch		Length of shank, inch(es)		Overall length of bur, inch(es)
	Min.	Max.	Min.	Max.	
AL	0.0919	0.0929	0.420	0.548	0.875 ± 0.010
S	0.0919	0.0929	1.100	1.410	1.750 ± 0.010

Class AL burs shall have shanks conforming to figure 1, and class S burs shall have shanks conforming to figure 2 for style, design, and dimensions. Class AL burs shall have a clean machined latch end free of imperfections such as burrs and slivers.

3.3.2 Head and neck. The head and neck of the burs shall be straight, symmetrical, and properly formed. The blades shall be uniformly cut and sharp. The head and neck dimensions and the number of blades for the burs, shall be as specified in table II for the applicable class, shape, and size number. The shape of the bur heads listed in table II shall be as shown in figures 3 through 9.

Table II. Bur head and neck data

Fig. No.	Class	Shape	Size Number	No. of blades (± 1)	Length of head inch /1	Diameter of head, inch /2	Diameter of neck, inch, max.
3	AL	Inter-proximal	6	18	0.197	0.0715	/3
4	AL	Round	200	20	-----	0.081	0.041
5	AL	Oval	218	20	0.038	0.081	0.044
6	AL	Bud	224	20	0.123	0.081	0.044
7	AL	Sugar-loaf	242	10	0.170	0.039	0.032
8	S	Round	200	20	-----	0.081	0.041
9	S	Bud	224	20	0.123	0.081	0.044

/1 The tolerance on the length of head of the AL interproximal bur shall be ± 0.008 inch. Head lengths for the other burs are minimums.

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/2 The head diameter shall be designated at the maximum cross-section on the head. The tolerance for the AL interproximal bur shall be ± 0.0035 inch. The tolerance for all the other burs shall be ± 0.002 inch.

/3 The AL interproximal bur shall have a neck length of 0.189 to 0.205 inch. The maximum neck diameter for this bur is not applicable.

3.4 Runout. The total indicated runout of the bur shall not exceed 0.003 inch, when tested as specified in 4.4.4.

3.5 Finish. The bur head shall have a finely machined finish and the blades shall be properly sharpened. The shank and neck shall have a smooth finish.

3.6 Identification marking. Each bur shank shall be stamped with the name or registered trademark of the manufacturer. Such marking shall be legible and permanent.

3.7 Workmanship. Workmanship shall be first class throughout. The burs shall be free from defects which detract from their appearance or may impair their serviceability, proper functioning and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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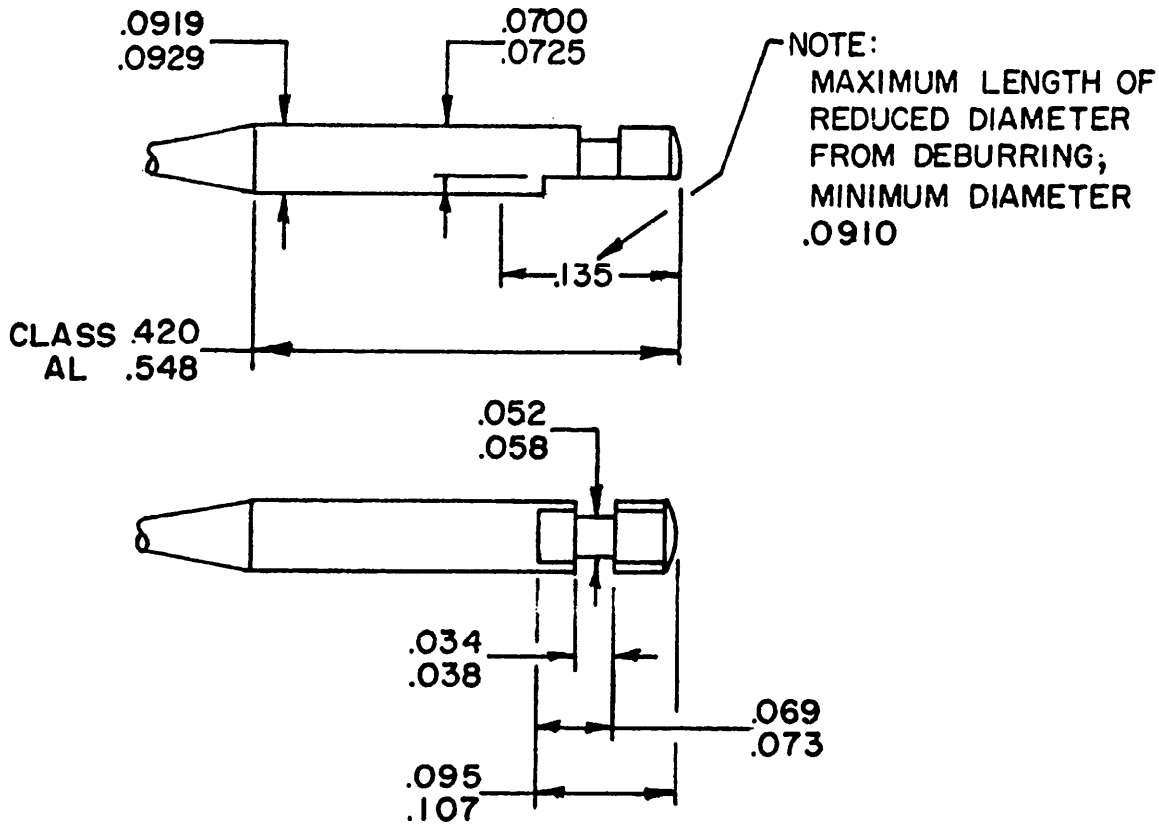


FIGURE 1. CLASSES AL

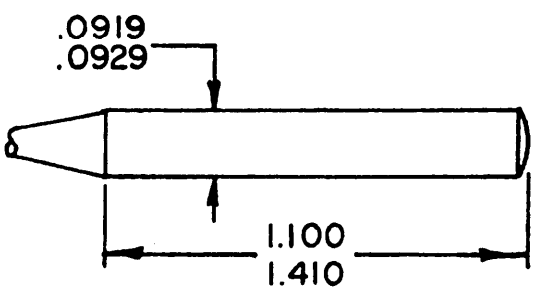


FIGURE 2. CLASS S

NOTE:
ALL DIMENSIONS ARE IN INCHES

DIMENSIONS OF SHANK END OF BUR

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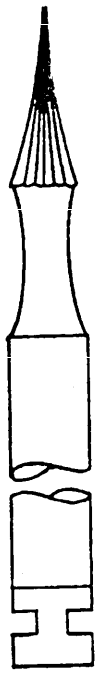


FIG 3

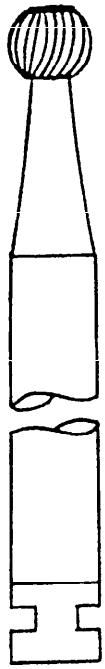


FIG 4

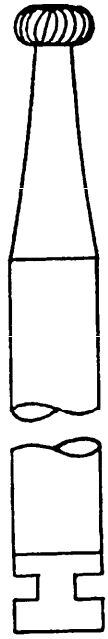


FIG 5

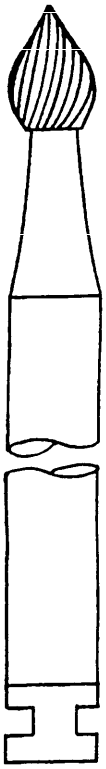


FIG 6

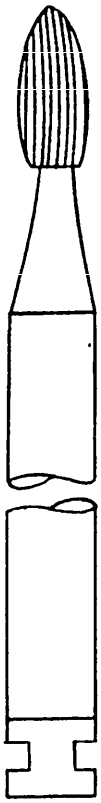


FIG 7

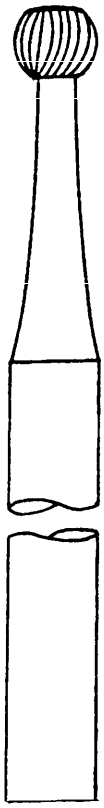


FIG 8



FIG 9

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection level and acceptable quality level (AQL) shall be as specified in table III.

Table III. Sampling for Examination

Type defect	Inspection level	AQL (percent defective)
Major	II	2.5
Minor	II	6.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified in table IV.

Table IV. Sampling for Tests

Test	Requirement	Test Procedure	Unit of product	Inspection level	Acceptance Number
Material	3.1	4.4.1	N/A /1	N/A /1	/2
Hardness:					
Scratch	3.2.1	4.4.2	1 bur	S-1	/2
Vickers	3.2.2	4.4.3	1 bur	S-1	/2
Runout	3.4	4.4.4	1 bur	S-4	2.5

/1 N/A Not applicable.
/2 The acceptance number shall be zero for all sample sizes.

4.2.3 For chemical analysis. A sufficient quantity of each lot or batch of raw material shall be selected for chemical analysis.

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4.3 Examination. The burs shall be examined for the possible defects listed in table V. The examination shall not be restricted to those defects shown in the table. Unless otherwise specified in table V, the defect listed shall apply to both the AL and S classes of burs contained in this specification.

Table V. Classification of defects

Category	Defect
Major:	
101	Burs not of shape and size specified.
102	Burs not of class specified.
103	Shank design and dimensions not as specified.
104	Design and dimensions of latch end of class AL burs not as specified.
105	Shapes and dimensions of head and neck not as specified.
106	Blades not properly cut; not sharp and free from nicks.
107	Finish of bur heads not as specified.
108	Latch end of class AL burs not clean machined.
Minor:	
201	Overall lengths not as specified.
202	Shank lengths not as specified.
203	Finish of necks and shanks not as specified.
204	Packaging, packing, and marking not as specified.
205	Marking on burs missing or improper.

4.3.1 Dimensions. Examination of the dimensions, as listed in 3.2, shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.3.2 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 30 power (30X) magnification.

4.4 Tests.

4.4.1 Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the burs shall be determined using method No. 111 or 112 of FED-STD-151.

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4.4.2 Scratch test. The bur shall be inserted into the proper handpiece, and the handpiece held so that it cannot be rotated. The bur shall be presented to a piece of plate glass testing material, with the handpiece held at an angle of five to ten degrees above horizontal. A pressure of approximately three pounds shall be used. The bur shall be dragged at right angles to its axis for a short pull against the glass. After the test has been completed, the bur head and plate glass shall be examined for compliance with 3.2.1.

4.4.3 Vickers test. The Vickers hardness test, if required, shall be made in accordance with method No. 244 of FED-STD-151.

4.4.4 Runout. The total indicated runout of the bur neck shall be measured to the nearest 0.0005 inch. The bur shall be held in a vee-block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the bur and shall be so positioned that the foot contacts the bur neck at the point of smallest diameter. The difference between the maximum and minimum readings for one full rotation of the bur shall be taken as the total indicated runout.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (6 burs), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	1152 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. The burs shall be treated with corrosion preventive conforming to grade 3 of MIL-C-16173. Six burs so treated shall be inserted into a plastic holder in such a manner that the burs will not contact one another. The burs shall be further protected with either a clear plastic cover which shall be secured in position by means of pressure sensitive tape, or a paperboard cover which shall be overwrapped with moisture-resistant cellulose acetate and heat sealed.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Class, shape, and size number required (see 1.2.1 and 1.2.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

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6.3 Supersession data. This specification includes the requirements for type II, finishing burs, of GG-B-821, dated 14 May 1948.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements .

SPECIFICATION: GG-B-001275(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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